

Basic Electrical & Electronics Engineering

Unit – 2 : Electrical Machines & Measuring Instruments

 Complete Theory

 Neat Diagrams

 Solved Numericals

 Exam Questions

 Quick Revision

UNIT 2

Syllabus at a Glance

Electrical Machines

- DC Motor – Construction & Working
- DC Generator – EMF Equation
- Single Phase Transformer
- Three Phase Induction Motor
- Alternator (Synchronous Generator)
- Applications of Electrical Machines

Measuring Instruments

- PMMC Instrument
- Moving Iron (MI) Instrument
- Wheatstone Bridge
- Comparison of PMMC vs MI
- Solved Numericals

 **EXAM TIP:** JNTUK exams frequently ask "Explain the construction and working principle of [machine] with a neat diagram." (10 marks). Each machine/instrument section in this document is structured to answer such questions directly.

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PART A – ELECTRICAL MACHINES

1. DC Motor Direct Current Motor

Definition: A DC Motor is an electromechanical device that converts **electrical energy (DC) into mechanical energy (rotation)**. It operates on the principle that a current-carrying conductor placed in a magnetic field experiences a mechanical force (Lorentz Force).

1.1 Construction of DC Motor

- **Yoke (Frame):** Outer cylindrical body made of cast iron or silicon steel; provides mechanical support and magnetic path.
- **Field Poles & Field Winding:** Electromagnets wound with copper wire; produce the main magnetic flux (Φ).
- **Armature Core:** Laminated silicon steel cylinder; reduces eddy current losses. Contains slots for armature winding.
- **Armature Winding:** Copper conductors placed in armature slots; carry current and experience force.
- **Commutator:** Cylindrical assembly of copper segments separated by mica; converts alternating torque into unidirectional torque.
- **Brushes:** Carbon brushes maintain sliding electrical contact with commutator; transfer current from external circuit to armature.
- **Shaft:** Transmits mechanical power to the load.

FIG 1.1 – DC MOTOR CROSS-SECTIONAL DIAGRAM

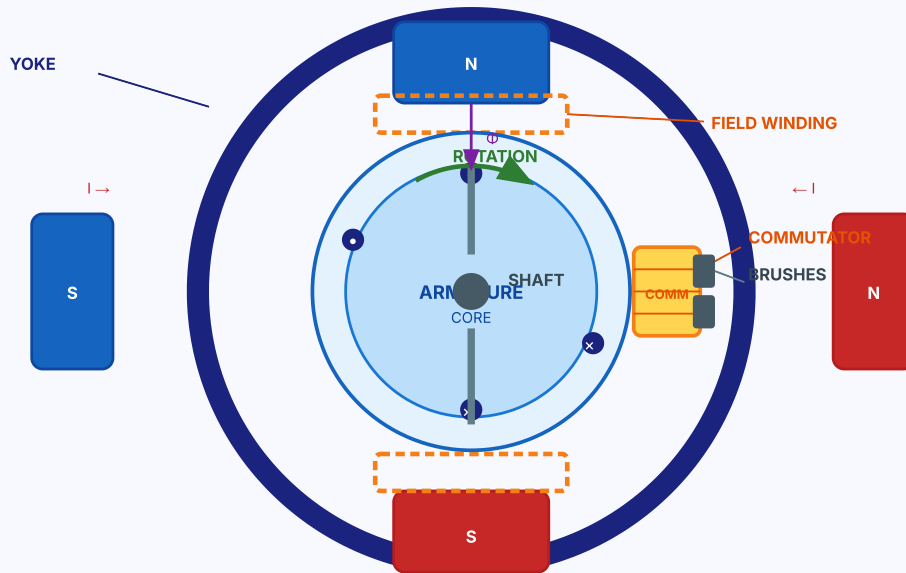


Fig 1.1: Cross-sectional view of a DC Motor showing armature, field poles, commutator, brushes and shaft

1.2 Working Principle of DC Motor

- 1 DC supply is connected to the armature through brushes and commutator.
- 2 Current flows through the armature conductors placed in the magnetic field created by field poles.
- 3 By **Lorentz Force Law**: $F = BIL$, each conductor experiences a force.
- 4 Using **Fleming's Left Hand Rule**: thumb $\rightarrow$ motion, index $\rightarrow$ field, middle $\rightarrow$ current direction.
- 5 Forces on opposite conductors act in the same rotational direction, creating **torque**.
- 6 The commutator ensures current direction in armature conductors reverses every half revolution, maintaining continuous rotation in one direction.

KEY FORMULAS – DC MOTOR

$$T = (\Phi Z I A) / (2\pi A) \quad [\text{Torque}]$$

$$E_b = V - I_a R_a \quad [\text{Back EMF}]$$

$$N \propto E_b / \Phi \quad [\text{Speed Relation}]$$

Where: P = poles, Φ = flux/pole (Wb), Z = conductors, I_a = armature current, R_a = armature resistance, V = supply voltage

✓ Advantages

- High starting torque
- Easy speed control
- Simple and robust construction
- Good overload capacity

✗ Disadvantages

- Requires DC supply
- Brushes wear and need maintenance
- Commutator causes sparking
- Not suitable for explosive environments

🔧 **Applications of DC Motor:** Electric traction (trains, trams), cranes & hoists, rolling mills, lifts/elevators, printing presses, electric vehicles (DC series motor), lathes and milling machines.

2. DC Generator Direct Current Generator

Definition: A DC Generator is an electromechanical device that converts **mechanical energy** into **electrical energy (DC)**. It works on the principle of **Faraday's Law of Electromagnetic**

Induction: an EMF is induced in a conductor whenever there is a change in the magnetic flux linked with it.

2.1 Construction of DC Generator

The construction of a DC Generator is identical to a DC Motor. Major parts:

- **Stator (Frame + Field Poles):** Fixed part; provides magnetic flux. Field poles wound with field winding excited by DC.
- **Armature:** Rotating part; made of laminated silicon steel. Armature winding placed in slots generates EMF.
- **Commutator:** Converts alternating induced EMF into unidirectional (DC) output at terminals.
- **Brushes & Slip Rings:** Carbon brushes collect current from rotating commutator and deliver to external circuit.
- **Output Terminals:** +ve and -ve terminals provide DC output.

FIG 2.1 – DC GENERATOR DIAGRAM

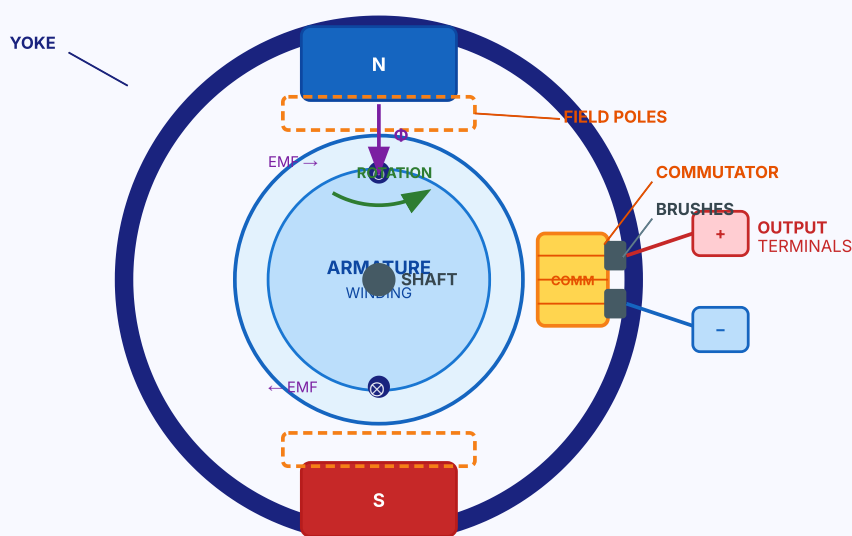


Fig 2.1: DC Generator – showing field poles, armature, commutator, brushes and DC output terminals

2.2 Working Principle of DC Generator

- 1 Armature is driven mechanically (prime mover: steam turbine, engine, etc.).
- 2 Armature conductors rotate in the magnetic field of field poles.
- 3 By **Faraday's Law**: $e = -d\Phi/dt$; EMF is induced in armature conductors.
- 4 **Fleming's Right Hand Rule** gives direction: thumb $\rightarrow$ motion, index $\rightarrow$ field, middle $\rightarrow$ induced EMF.
- 5 Induced EMF is alternating (AC) inside armature, but commutator + brushes rectify it to DC at output.
- 6 DC output is collected at +ve and -ve output terminals.

EMF EQUATION OF DC GENERATOR

$$E = (P \times \Phi \times Z \times N) / (60 \times A)$$

P = Number of poles | **Φ** = Flux per pole (Wb) | **Z** = Total armature conductors

N = Speed (RPM) | **A** = Parallel paths (A=P for lap winding; A=2 for wave winding)

✓ Advantages

- Wide voltage range possible
- Good for battery charging
- Self-excited types available
- Reliable for electroplating

✗ Disadvantages

- Commutator maintenance required
- Commutation sparking issues
- Not suitable for high voltage
- Brush replacement periodically needed

🔧 **Applications of DC Generator:** Battery charging, arc welding, electroplating, excitation of large AC generators, field supply for DC motors, electrolysis processes.

3. Single Phase Transformer Electromagnetic Energy Transfer Device

Definition: A transformer is a static electromagnetic device that transfers electrical energy from one circuit to another through **electromagnetic induction without changing the frequency**. It can step up or step down AC voltage and current levels.

Principle: Based on **Faraday's Law of Mutual Induction** – when AC flows in primary winding, the alternating flux links with the secondary winding and induces an EMF.

3.1 Construction of Single Phase Transformer

- **Magnetic Core:** Made of high-grade silicon steel laminations (0.3–0.5 mm thick). Reduces eddy current losses. Provides low reluctance path for magnetic flux. Two types: Core type and Shell type.
- **Primary Winding:** Connected to AC supply. Has N_1 turns. Draws magnetizing current and creates alternating flux.
- **Secondary Winding:** Connected to load. Has N_2 turns. EMF induced by mutual induction; delivers power to load.
- **Insulation:** Paper/varnish insulation between layers; prevents electrical contact between windings.
- **Tank & Oil:** Large transformers immersed in insulating oil for cooling and insulation.

FIG 3.1 – SINGLE PHASE TRANSFORMER DIAGRAM (CORE TYPE)

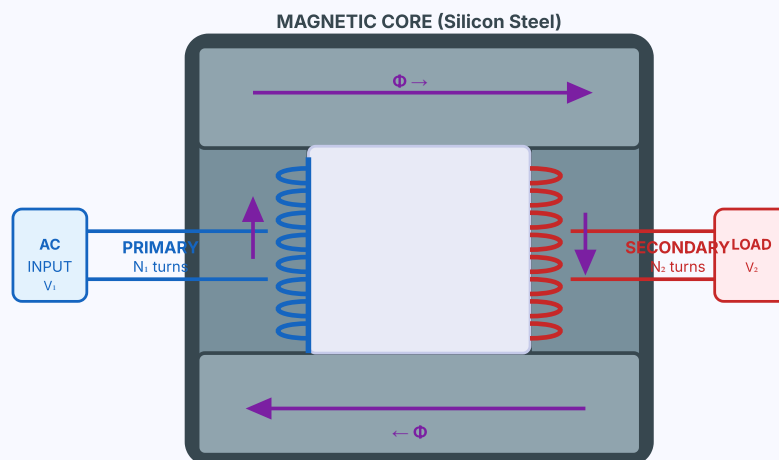


Fig 3.1: Single Phase Transformer – Primary winding (blue), Secondary winding (red), Magnetic core (grey), Flux path Φ (purple arrows)

3.2 Working Principle

- 1 AC supply (V_1) connected to primary winding; alternating current I_1 flows.
- 2 Alternating current creates alternating magnetic flux Φ in the core: $\Phi = \Phi_m \sin(\omega t)$
- 3 Flux links with secondary winding (mutual induction). EMF induced: $e_2 = -N_2 d\Phi/dt$
- 4 Turns ratio determines voltage transformation: $V_1/V_2 = N_1/N_2 = I_2/I_1 = a$ (transformation ratio)
- 5 If $N_2 > N_1 \rightarrow$ Step-up transformer ($V_2 > V_1$). If $N_2 < N_1 \rightarrow$ Step-down transformer ($V_2 < V_1$).

EMF EQUATION OF TRANSFORMER

$$E_1 = 4.44 \times f \times N_1 \times \Phi_m$$

$$E_2 = 4.44 \times f \times N_2 \times \Phi_m$$

$$\text{Turns Ratio: } a = N_1/N_2 = V_1/V_2 = I_2/I_1$$

f = frequency (Hz) | N_1, N_2 = turns | Φ_m = max flux (Wb) | $4.44 = 4 \times (\sqrt{2}/2)\pi$ = form factor $\times \pi$

3.3 Transformer Losses

Loss Type	Cause	Location	How to Reduce
Iron/Core Loss (constant)	Hysteresis + Eddy currents	Magnetic Core	CRGO steel, thin laminations
Copper/Ohmic Loss (variable)	I^2R heating of windings	Primary & Secondary	Thick conductors, low resistance
Stray Loss	Leakage flux, capacitance	Tank, windings	Proper design & shielding

TRANSFORMER EFFICIENCY

$$\eta = (\text{Output Power}) / (\text{Output Power} + \text{Core Loss} + \text{Copper Loss}) \times 100\%$$

$$\eta = (V_2 I_2 \cos\phi) / (V_2 I_2 \cos\phi + W_c + W_i) \times 100\%$$

W_c = copper loss (W), W_i = iron loss (W), $\cos\phi$ = power factor of load

✓ Advantages

- Very high efficiency (98–99%)
- No moving parts (static)
- Low maintenance
- Works for wide voltage range

✗ Disadvantages

- Works only on AC (not DC)
- Large and heavy for high rating
- Insulation oil is flammable
- Core and copper losses exist

🔧 **Applications:** Power transmission & distribution, welding transformers, isolation transformers, audio amplifiers (impedance matching), voltage stabilizers, mobile phone chargers (SMPS uses transformer principle).

4. Three Phase Induction Motor Most Widely Used Industrial Motor

Definition: A three-phase induction motor is an AC motor in which electric power is transferred from the stator (primary) to the rotor (secondary) by **electromagnetic induction**, similar to transformer action. The rotor receives no direct electrical connection – power is transferred entirely by induction.

Principle: When 3-phase AC supply is given to the stator winding, a **Rotating Magnetic Field (RMF)** is produced. This RMF induces current in rotor conductors, which in turn produces a rotor magnetic field, and the interaction of both fields produces the torque that rotates the rotor.

4.1 Construction

Stator

- Outer stationary part
- Laminated silicon steel frame
- Contains 3-phase winding (120° apart)
- Connected to 3-phase AC supply
- Creates rotating magnetic field

Rotor (Two Types)

- **Squirrel Cage Rotor:** Short-circuited copper/aluminium bars in slots; simple, robust, no external connections.
- **Wound/Slip Ring Rotor:** 3-phase winding connected via slip rings; allows external resistance for speed control.

FIG 4.1 – THREE PHASE INDUCTION MOTOR (CROSS SECTION)

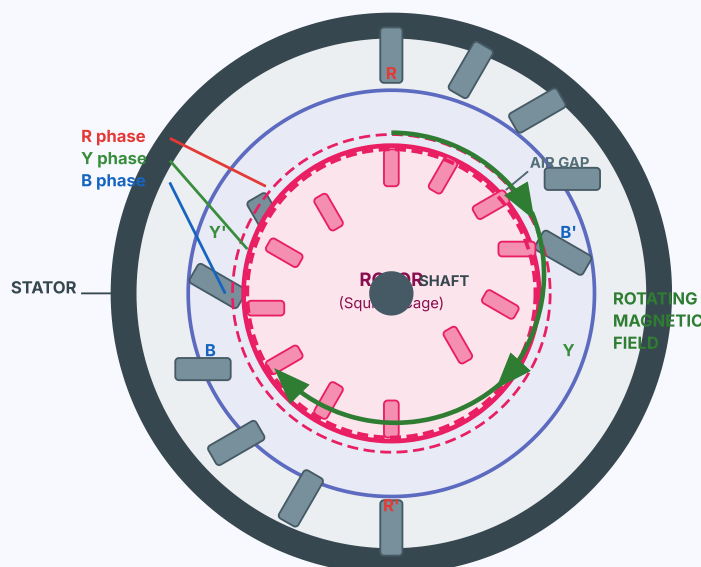


Fig 4.1: 3-Phase Induction Motor – Stator (3-phase winding R,Y,B), Squirrel Cage Rotor, Rotating Magnetic Field (green arrows)

4.2 Working Principle (Step-by-Step)

- 1 3-phase AC supply (120° phase displaced) applied to stator winding produces a **Rotating Magnetic Field (RMF)** at synchronous speed: $N_s = 120f/P$.
- 2 RMF cuts through stationary rotor conductors $\rightarrow$ EMF induced in rotor (by Faraday's Law).
- 3 Since rotor conductors are short-circuited (in squirrel cage), induced EMF causes rotor current to flow.
- 4 Rotor current-carrying conductors in RMF experience force ($F = BIL$), creating torque on rotor.
- 5 Rotor starts rotating in direction of RMF but always slightly slower than RMF (slip $s > 0$).
- 6 If rotor runs at synchronous speed $\rightarrow$ no relative motion $\rightarrow$ no induced EMF $\rightarrow$ no torque $\rightarrow$ rotor can never reach N_s . Hence the name "**Induction Motor**".

KEY FORMULAS – 3-PHASE INDUCTION MOTOR

$$N_s = 120f / P \quad [\text{Synchronous Speed, RPM}]$$

$$\text{Slip: } s = (N_s - N_r) / N_s$$

$$N_r = N_s(1 - s) \quad [\text{Rotor Speed}]$$

$$\text{Rotor frequency: } f_r = s \times f$$

f = supply frequency (Hz) | P = number of poles | N_r = actual rotor speed | s = slip ($0 < s < 1$)

✓ Advantages

- Robust & simple construction
- Self-starting (no starter needed)
- Low cost & maintenance
- High efficiency (85–95%)

✗ Disadvantages

- Speed control difficult
- Low power factor at light load
- High inrush starting current
- Speed varies with load

 **Applications:** Industrial pumps, compressors, fans, blowers, conveyor belts, machine tools, crushers, elevators, cranes. Most widely used motor in industry (>90% of motors are induction type).

5. Alternator (Synchronous Generator) AC Generator

Definition: An alternator (synchronous generator) is a machine that converts **mechanical energy into 3-phase alternating electrical energy**. It runs at synchronous speed $N_s = 120f/P$, and the output frequency is fixed by this speed. It is the primary source of electrical power generation in power plants.

5.1 Construction of Alternator

Stator (Armature)

- Outer stationary part
- Laminated silicon steel core
- 3-phase armature winding placed in slots
- Generates 3-phase output voltage
- Output terminals: R, Y, B phases + Neutral

Rotor (Field)

- Inner rotating part; driven by prime mover
- **Salient Pole:** Used in slow-speed alternators (hydro), distinct projecting poles.
- **Cylindrical/Round Rotor:** Used in high-speed turbine-driven alternators (thermal/nuclear).
- DC excitation via slip rings

FIG 5.1 – ALTERNATOR (SYNCHRONOUS GENERATOR) DIAGRAM

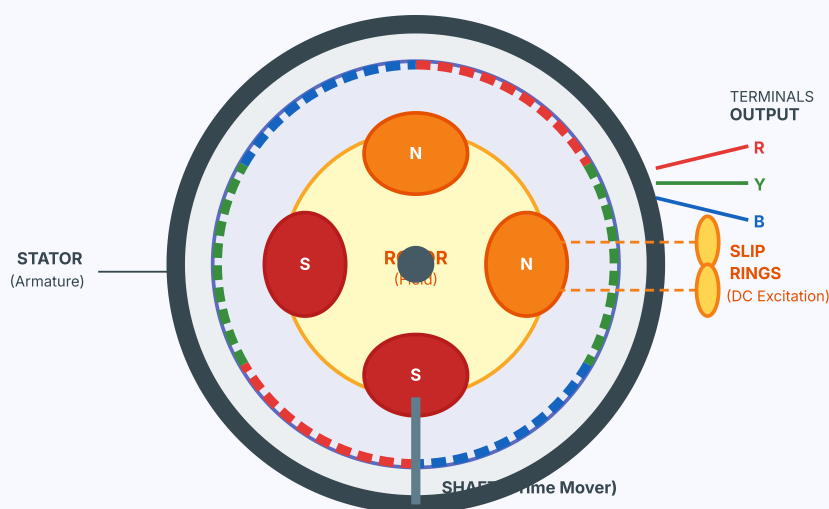


Fig 5.1: Alternator – Stator armature (3-phase winding), Salient pole Rotor (N/S poles), Slip rings (DC excitation), Output terminals (R,Y,B)

5.2 Working Principle

- 1 Rotor (field system) is driven by a prime mover (steam turbine, hydro turbine) at synchronous speed N_s .
- 2 DC excitation is provided to rotor field winding through slip rings → creates stationary rotor field poles (N and S alternating).
- 3 Rotating rotor poles sweep past the stationary 3-phase stator armature windings → magnetic flux cuts stator conductors.
- 4 By Faraday's Law: EMF induced in each stator phase winding. Since poles alternate N-S-N-S, induced EMF is **alternating (AC)**.
- 5 Three stator windings (120° spatially apart) produce 3-phase EMF (120° time phase apart).
- 6 Output frequency: $f = P \times N / 120$ (strictly tied to rotor speed).

EMF EQUATION OF ALTERNATOR

$$E_{ph} = 4.44 \times K_w \times f \times \Phi \times T_{ph}$$

$$f = \frac{PN}{120} \quad [\text{Output Frequency}]$$

K_w = winding factor (0.85–0.95) | T_{ph} = turns per phase | Φ = flux per pole (Wb) | P = poles | N = rotor speed (RPM)

✓ Advantages

- High voltage generation possible
- 3-phase output directly usable
- High efficiency & reliability
- Frequency precisely controlled by speed

✗ Disadvantages

- Requires constant speed regulation
- Slip rings and brushes need maintenance
- Parallel operation needs synchronization
- Costly excitation system needed

🔧 **Applications:** Thermal power plants (coupled with steam turbines), hydroelectric stations (water turbines), diesel power plants (diesel engines), aircraft alternators, wind turbines (DFIG type), standby/emergency power generators.

SECTION 6

Applications of Electrical Machines – Summary Table

Machine	Type	Key Applications	Key Feature
DC Motor	Series, Shunt, Compound	Electric traction, cranes, lifts, rolling mills, printing press, electric vehicles	High starting torque (Series)
DC Generator	Self/Separately Excited	Battery charging, arc welding, electroplating, excitation of AC generators	Variable DC voltage output
Transformer	Step-up / Step-down	Power T&D, welding, isolation, instrument transformers, SMPS	Static; ~99% efficiency
3-Ph Induction Motor	Squirrel cage, Slip ring	Pumps, fans, compressors, conveyors, machine tools, crushers, elevators	Robust, self-starting, cheap
Alternator	Salient pole / Cylindrical	Thermal, hydro, nuclear, diesel power plants; wind turbines; standby generators	Primary 3-phase power source

 **EXAM TIP:** Questions like "List the applications of DC motor" or "Compare DC motor and DC generator" are frequently asked for 5 marks in JNTUK. This table covers both types of questions.



PART B – MEASURING INSTRUMENTS

7. PMMC Instrument Permanent Magnet Moving Coil

Definition: A PMMC (Permanent Magnet Moving Coil) instrument is an electrical measuring device that uses the interaction between the magnetic field of a permanent magnet and a current-carrying coil to deflect a pointer and indicate the measured quantity. It is the most accurate and sensitive type of instrument used for DC measurement.

Principle: When a current-carrying conductor is placed in a magnetic field, it experiences a force ($F = BIL$). This force creates a deflecting torque proportional to current.

7.1 Construction of PMMC

- **Permanent Magnet:** Horseshoe-shaped or circular permanent magnet (Alnico alloy); provides strong uniform radial magnetic field in the air gap.
- **Pole Pieces & Cylindrical Core:** Pole pieces are curved; iron cylindrical core placed between them creates uniform radial field, ensuring linear scale.
- **Moving Coil:** Rectangular coil of fine copper wire wound on aluminium former; free to rotate in air gap; carries the measuring current.
- **Pointer:** Lightweight aluminium pointer attached to coil spindle; deflects over scale.
- **Control Springs (Hair Springs):** Two spiral hairsprings (upper and lower) provide controlling torque proportional to deflection; also serve as current leads to coil.
- **Damping:** Provided by induced currents in aluminium former (eddy current damping).
- **Jewel Bearings:** Pivot bearings support the coil spindle; minimize friction.
- **Scale:** Uniform (linear) scale since $T_d \propto I$.

FIG 7.1 – PMMC INSTRUMENT DIAGRAM

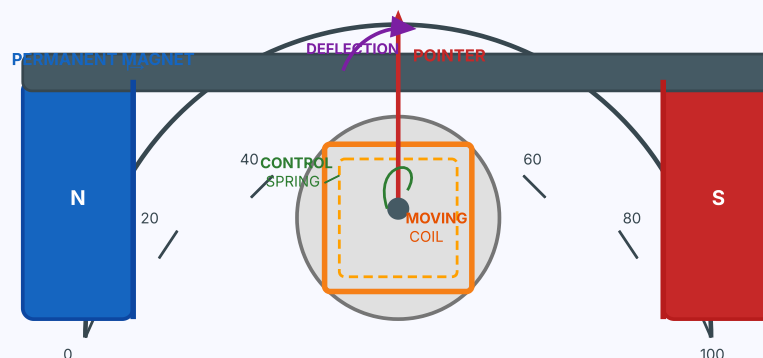


Fig 7.1: PMMC Instrument – Permanent Magnet (N pole blue, S pole red), Moving Coil (orange), Pointer (red), Control Spring (green), Deflection arrow (purple)

DEFLECTING TORQUE – PMMC

$$T_d = B \times I \times l \times w \times N = BINA \text{ (where } A = l \times w = \text{coil area)}$$

$$\text{At equilibrium: } T_d = T_c \rightarrow BINA = K\theta \rightarrow \theta = (BINA)/K \propto I$$

θ = deflection (rad) | K = spring constant | I = current | N = turns | A = coil area | B = flux density

8. Moving Iron (MI) Instrument Attraction & Repulsion Type

Definition: A Moving Iron instrument works on the principle that a piece of soft iron placed near a current-carrying coil is **attracted into a stronger magnetic field**. Unlike PMMC, it works for both AC and DC. The deflecting force is proportional to the square of the current.

Two Types: (1) **Attraction Type** – iron piece attracted into coil | (2) **Repulsion Type** – two iron pieces (one fixed, one moving) repel each other (both magnetised in same direction by coil current).

8.1 Construction (Repulsion Type)

- **Fixed Coil:** Non-inductive coil wound on cylindrical former; carries the measuring current. Coil remains stationary.
- **Fixed Iron Vane:** Soft iron strip fixed inside the coil, magnetised by coil current.
- **Moving Iron Vane:** Soft iron strip attached to the spindle; free to move; also magnetised by coil current in same direction → repulsion between the two iron pieces.
- **Pointer:** Attached to the spindle of moving iron vane; moves over scale.
- **Control Spring:** Provides controlling torque.
- **Air Friction Damping:** Aluminium disc in a curved air chamber provides damping (piston type).

FIG 8.1 – MOVING IRON (REPULSION TYPE) INSTRUMENT

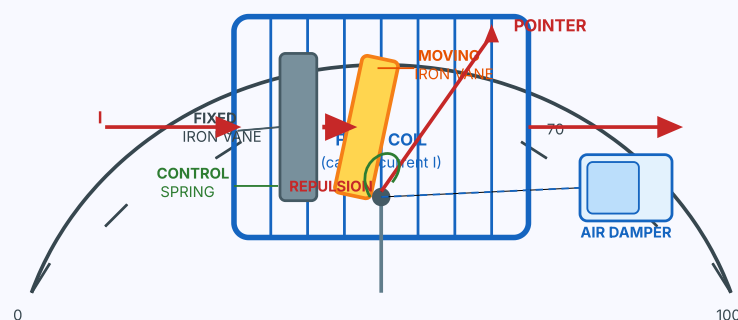


Fig 8.1: Moving Iron (Repulsion Type) – Fixed coil (blue), Fixed iron vane (grey), Moving iron vane (yellow), Repulsion force (red arrow), Control spring (green), Pointer

DEFLECTING TORQUE – MOVING IRON

$$T_d = (1/2) \times I^2 \times (dL/d\theta)$$

$$\text{At equilibrium: } (1/2) \times I^2 \times (dL/d\theta) = K\theta \rightarrow \theta \propto I^2$$

θ = deflection | L = inductance of coil | K = spring constant | Scale is NON-UNIFORM (cramped at low end)

💡 **EXAM TIP:** MI instruments measure BOTH AC and DC (since $T_d \propto I^2$; squaring removes sign). PMMC measures DC only. This is a frequently asked 2-mark difference!

8.2 Comparison: PMMC vs Moving Iron Instrument

Parameter	PMMC	Moving Iron (MI)
Working Principle	Current in field of permanent magnet	Attraction/Repulsion of iron by coil
Supply Type	DC only	Both AC and DC
Scale	Uniform (linear)	Non-uniform (cramped at start, I^2)
Accuracy	High (Class 0.1 to 1)	Moderate (Class 1 to 2.5)
Sensitivity	Very high	Low
Friction Effect	More sensitive to friction	Less sensitive to friction
Hysteresis Error	None	Present (due to iron)
Damping	Eddy current (aluminium former)	Air friction (piston type)
Cost	Higher	Lower
Use	Galvanometers, ammeters, voltmeters (DC)	AC/DC ammeters, voltmeters

9. Wheatstone Bridge Precision Resistance Measurement

Definition: A Wheatstone Bridge is a circuit used to **measure an unknown resistance precisely** by balancing two legs of a bridge circuit. When the bridge is balanced, no current flows through the galvanometer, and the unknown resistance R_x can be calculated from the known resistances.

Principle: At balance condition, the ratio of resistances in one pair of arms equals the ratio in the other pair: $R_1/R_2 = R_3/R_x$

9.1 Construction / Circuit

- **Four Arms:** Four resistors (R_1 , R_2 , R_3 , R_x) form a quadrilateral diamond bridge circuit.
- **Battery (EMF Source):** Connected between junctions A and C (supply diagonal).
- **Galvanometer (G):** Connected between junctions B and D (detector diagonal); sensitive current detector.
- **R_1 , R_2 :** Known ratio arms (standard resistors).
- **R_3 :** Variable/standard arm (adjustable known resistance).
- **R_x :** Unknown resistance to be measured.

FIG 9.1 – WHEATSTONE BRIDGE CIRCUIT DIAGRAM

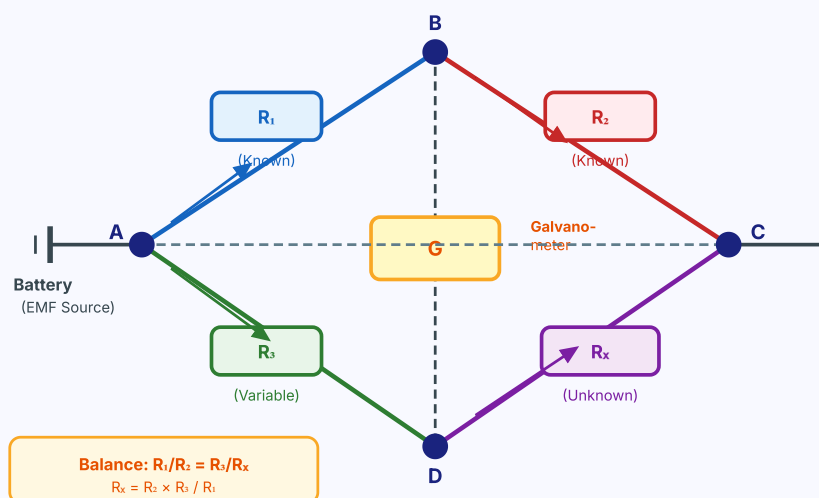


Fig 9.1: Wheatstone Bridge – Arms R_1, R_2 (ratio arms), R_3 (standard), R_x (unknown), Galvanometer G at BD , Battery at AC , Current flow arrows shown

9.2 Working Principle

- 1 Battery is connected across junctions A and C; Galvanometer across B and D.
- 2 R_3 is adjusted until the galvanometer reads zero (null deflection).
- 3 At null/balance condition: current through galvanometer = 0, meaning potential at B = potential at D.
- 4 Using Kirchhoff's Voltage Law at balance: $I_1 R_1 = I_2 R_2$ and $I_1 R_3 = I_2 R_x$
- 5 Dividing: $R_1/R_3 = R_2/R_x \rightarrow \mathbf{R_x = R_2 \times R_3 / R_1}$

WHEATSTONE BRIDGE – BALANCE CONDITION

$$R_1 \times R_x = R_2 \times R_3$$

$$\therefore R_x = (R_2 \times R_3) / R_1$$

At balance: Galvanometer current $I_G = 0$ | Potential at B = Potential at D | Bridge is in null deflection state

✓ Advantages

- Very accurate null method
- Independent of supply voltage
- Simple circuit; easy to use
- Wide range of resistance measurement

✗ Disadvantages / Limitations

- Not suitable for very low (contact resistance) or very high resistance
- Error due to contact and lead resistance
- Heating of resistors by battery current
- Galvanometer sensitivity affects accuracy

🔧 **Applications:** Precision resistance measurement in laboratories, strain gauge measurement (bridge circuit), temperature sensors (RTD), biosensors, Kelvin double bridge (for very low R), AC bridges (impedance measurement).

 **EXAM TIP:** Numericals carry 5–10 marks. Always write: Given → Find → Formula → Substitution → Answer. Show units at each step!

Problem 1 (DC Generator – EMF Equation)

A 6-pole DC generator has a lap-wound armature with 360 conductors. The flux per pole is 0.04 Wb and the generator runs at 900 RPM. Calculate the generated EMF.

Given: $P = 6$, $Z = 360$, $\Phi = 0.04$ Wb, $N = 900$ RPM, Lap winding $\rightarrow A = P = 6$

Formula: $E = \frac{P\Phi ZN}{60A}$

Substitution: $E = \frac{(6 \times 0.04 \times 360 \times 900)}{(60 \times 6)}$

$$E = \frac{(6 \times 0.04 \times 360 \times 900)}{360}$$

$$E = 77760 / 360$$

 **EMF (E) = 216 V**

Problem 2 (DC Generator – Wave Winding)

A 4-pole DC generator with wave winding has 500 conductors. Flux per pole = 0.02 Wb, speed = 1500 RPM. Find the generated EMF.

Given: $P = 4$, $Z = 500$, $\Phi = 0.02$ Wb, $N = 1500$ RPM, Wave winding $\rightarrow A = 2$

Formula: $E = \frac{P\Phi ZN}{60A}$

Substitution: $E = \frac{(4 \times 0.02 \times 500 \times 1500)}{(60 \times 2)}$

$$E = 60000 / 120$$

 **EMF (E) = 500 V**

Problem 3 (Transformer – EMF Equation)

A single-phase transformer has 200 primary turns and 1000 secondary turns. The core has a maximum flux of 0.05 Wb and frequency = 50 Hz. Find primary and secondary EMFs.

Given: $N_1 = 200$, $N_2 = 1000$, $\Phi_m = 0.05$ Wb, $f = 50$ Hz

Formula: $E_1 = 4.44 \times f \times N_1 \times \Phi_m$ $E_2 = 4.44 \times f \times N_2 \times \Phi_m$

$$E_1: E_1 = 4.44 \times 50 \times 200 \times 0.05 = 4.44 \times 500 = 2220 \text{ V}$$

$$E_2: E_2 = 4.44 \times 50 \times 1000 \times 0.05 = 4.44 \times 2500 = 11100 \text{ V}$$

 **$E_1 = 2220$ V (Primary EMF) | $E_2 = 11,100$ V (Secondary EMF) | Turns Ratio = 1:5 (Step-up)**

📌 Problem 4 (Transformer – Efficiency)

A 10 kVA transformer has iron losses of 150 W and full-load copper losses of 250 W. Calculate efficiency at full-load, unity power factor.

Given: $S = 10 \text{ kVA} = 10000 \text{ VA}$, $W_i = 150 \text{ W}$, $W_c = 250 \text{ W}$, $\cos \phi = 1$

Output Power: $P_{\text{out}} = S \times \cos \phi = 10000 \times 1 = 10000 \text{ W}$

Total Losses: $P_{\text{loss}} = W_i + W_c = 150 + 250 = 400 \text{ W}$

Formula: $\eta = P_{\text{out}} / (P_{\text{out}} + P_{\text{loss}}) \times 100\%$

$\eta = 10000 / (10000 + 400) \times 100 = 10000 / 10400 \times 100$

✅ Efficiency $\eta = 96.15\%$

📌 Problem 5 (3-Phase Induction Motor – Slip)

A 4-pole, 3-phase induction motor is connected to a 50 Hz supply. The rotor speed is 1440 RPM. Calculate: (a) Synchronous speed, (b) Slip, (c) Rotor frequency.

Given: $P = 4$, $f = 50 \text{ Hz}$, $N_r = 1440 \text{ RPM}$

(a) $N_s = 120f/P = 120 \times 50 / 4 = 6000/4 = 1500 \text{ RPM}$

(b) Slip: $s = (N_s - N_r)/N_s = (1500 - 1440)/1500 = 60/1500 = 0.04 \text{ (or } 4\%)$

(c) Rotor frequency: $f_r = s \times f = 0.04 \times 50 = 2 \text{ Hz}$

✅ $N_s = 1500 \text{ RPM}$ | Slip $s = 4\%$ | Rotor frequency $f_r = 2 \text{ Hz}$

📌 Problem 6 (Wheatstone Bridge – Unknown Resistance)

In a Wheatstone bridge, $R_1 = 100 \Omega$, $R_2 = 200 \Omega$, $R_3 = 150 \Omega$. At balance condition, find the unknown resistance R_x .

Given: $R_1 = 100 \Omega$, $R_2 = 200 \Omega$, $R_3 = 150 \Omega$

Balance Condition: $R_1 \times R_x = R_2 \times R_3$

Formula: $R_x = (R_2 \times R_3) / R_1$

$R_x = (200 \times 150) / 100 = 30000 / 100 = 300 \Omega$

✅ Unknown Resistance $R_x = 300 \Omega$

📌 Problem 7 (Alternator – Frequency and EMF)

A 6-pole alternator rotates at 1000 RPM. Each phase has 120 turns, winding factor $K_w = 0.96$, flux per pole = 0.05 Wb. Find (a) frequency and (b) phase EMF.

Given: $P = 6$, $N = 1000 \text{ RPM}$, $T_{ph} = 120$, $K_w = 0.96$, $\Phi = 0.05 \text{ Wb}$

(a) $f = PN/120 = 6 \times 1000 / 120 = 50 \text{ Hz}$

(b) $E_{ph} = 4.44 \times K_w \times f \times \Phi \times T_{ph}$

$E_{ph} = 4.44 \times 0.96 \times 50 \times 0.05 \times 120 = 4.44 \times 0.96 \times 300 = 1278.7 \text{ V}$

✅ Frequency $f = 50 \text{ Hz}$ | Phase EMF $E_{ph} \approx 1278.7 \text{ V}$

 **Pattern:** JNTUK R23 typically has: Unit-wise 10-mark questions (choose 1 of 2) and 5-mark questions (choose 2 of 4). Questions marked ★ are most frequently repeated.

Section A – 10 Mark Questions (Long Answer Type)

Q1 Explain the construction and working principle of a DC Generator with a neat diagram. Derive the EMF equation. **10 Marks ★★★**

Q2 Describe the construction and working of a Single Phase Transformer. Derive the EMF equation and explain the losses. **10 Marks ★★★**

Q3 Explain the working principle of a Three-Phase Induction Motor with a neat diagram. What is slip? Explain the significance of slip. **10 Marks ★★★**

Q4 Describe the construction and working of an Alternator (Synchronous Generator) with a neat diagram. Derive the EMF equation. **10 Marks ★★**

Q5 Explain with a neat diagram the construction and working of a DC Motor. State the types and applications. **10 Marks ★★**

Section B – 5 Mark Questions (Short Answer Type)

Q6 Explain the working principle of PMMC instrument with a neat diagram. State its advantages over Moving Iron instruments. **5 Marks ★★★**

Q7 With a neat circuit diagram, explain the working of a Wheatstone Bridge. Derive the balance condition. **5 Marks ★★★**

Q8 Explain the construction and working of a Moving Iron (repulsion type) instrument with a neat diagram. **5 Marks ★★**

Q9 Compare PMMC and Moving Iron instruments with respect to principle, accuracy, scale, supply type and applications. **5 Marks ★★**

Q10 List and explain the applications of: (a) DC Motor, (b) DC Generator, (c) Transformer, (d) 3-phase Induction Motor. **5 Marks ★★**

Q11 A 4-pole DC generator with lap winding has 480 conductors, flux per pole = 0.05 Wb, speed = 600 RPM. Find the generated EMF. [Ans: 240 V]

5 Marks ★★

Q12 State and explain Fleming's Left Hand Rule and Right Hand Rule. How are they applied to DC Motor and DC Generator respectively?

5 Marks ★★

Q13 Why does a 3-phase induction motor not run at synchronous speed? Explain the concept of slip with formula.

5 Marks ★

Q14 In a Wheatstone bridge, $R_1 = 50 \Omega$, $R_2 = 100 \Omega$, $R_3 = 75 \Omega$. Find R_x at balanced condition. Verify by balance equation.

5 Marks ★

Q15 Explain the types of losses in a transformer. How is efficiency of a transformer calculated? At what load condition is the efficiency maximum?

5 Marks ★

Section C – 2 Mark Questions (Very Short Answer)

Q16 State Fleming's Left Hand Rule. 2M

Q17 What is the EMF equation of a DC generator? 2M

Q18 Define slip in an induction motor. 2M

Q19 Why does PMMC work only on DC? 2M

Q20 What is the turns ratio of a transformer? 2M

Q21 State the balance condition of Wheatstone bridge. 2M

Q22 What is a commutator in a DC machine? 2M

Q23 Why are MI instruments preferred for AC measurements? 2M

Q24 What are the types of rotors in induction motor? 2M

Q25 Define synchronous speed. 2M

QUICK REVISION SHEET – Unit 2

All Key Formulas | Important Points | Exam Mnemonics



All Key Formulas

DC Generator EMF

$$E = P\Phi ZN / (60A)$$

DC Motor Back EMF

$$E_b = V - I_a R_a$$

DC Motor Speed

$$N \propto E_b / \Phi$$

DC Motor Torque

$$T = P\Phi Z I_a / (2\pi A)$$

Transformer EMF (Primary)

$$E_1 = 4.44 f N_1 \Phi_m$$

Transformer EMF (Secondary)

$$E_2 = 4.44 f N_2 \Phi_m$$

Turns Ratio

$$a = N_1/N_2 = V_1/V_2 = I_2/I_1$$

Transformer Efficiency

$$\eta = V_2 I_2 \cos\phi / (V_2 I_2 \cos\phi + W_c + W_i) \times 100\%$$

Synchronous Speed

$$N_s = 120f / P \text{ (RPM)}$$

Slip

$$s = (N_s - N_r) / N_s$$

Rotor Speed

$$N_r = N_s(1 - s)$$

Rotor Frequency

$$f_r = s \times f$$

Alternator Frequency

$$f = PN / 120$$

Alternator Phase EMF

$$E_{ph} = 4.44 K_w f \Phi T_{ph}$$

PMMC Deflection

$$\theta = BINA/K \ (\propto I, \text{ linear})$$

MI Deflection

$$\theta \propto I^2 \text{ (non-uniform scale)}$$

Wheatstone Bridge Balance

$$R_1 R_X = R_2 R_3$$

Unknown Resistance

$$R_X = (R_2 \times R_3) / R_1$$



Must-Know Key Points

Electrical Machines

- **DC Motor:** Electrical → Mechanical | Fleming's Left Hand Rule
- **DC Generator:** Mechanical → Electrical | Fleming's Right Hand Rule
- **Transformer:** AC only (not DC) | Static (no moving parts)
- **Induction Motor:** Rotor never reaches N_s ($s > 0$ always)
- **Squirrel Cage:** Simple, robust, no external resistance
- **Wound Rotor:** External resistance → speed control possible
- **Alternator:** Field rotates, Armature stationary (practical type)
- **Slip rings:** Used in Alternator (DC excitation) & Wound rotor IM
- **Commutator:** Used in DC machines only
- **RMF:** Produced by 3-phase supply in IM & Alternator stator

Measuring Instruments

- **PMMC:** DC only | Uniform scale | Eddy current damping | High accuracy
- **MI:** AC & DC | Non-uniform scale | Air friction damping | Hysteresis error
- **Wheatstone Bridge:** Null/balance method | Mid-range R best
- **Galvanometer = 0** at bridge balance (no current through G)
- **PMMC principle:** $F = BIL$ (Lorentz force)
- **MI principle:** Attraction/Repulsion of iron by magnetic field
- PMMC → Scale: Uniform ($\theta \propto I$)
- MI → Scale: Non-uniform, cramped at start ($\theta \propto I^2$)
- Wheatstone: Best for medium resistance ($1\ \Omega$ to $100\ k\Omega$)

Memory Aids & Exam Mnemonics

- **"FBI" Rule → Fleming's Left Hand:** Force (Thumb) – B field (Index) – Internal current (Middle) → for DC Motor
- **"FBE" Rule → Fleming's Right Hand:** Force/Motion (Thumb) – B field (Index) – EMF induced (Middle) → for DC Generator
- **Transformer "4.44":** $4.44 = 4 \times \sqrt{2}/2 \times \pi = 4 \times 1.11$ — form factor contribution to EMF equation
- **Induction Motor: "SLIP = GAP"** – Slip ensures relative motion, which is the GAP between N_s and N_r that keeps the motor working!
- **Wheatstone balance: "Products of Opposite Arms Equal"** – $R_1 \times R_X = R_2 \times R_3$
- **PMMC vs MI in one line:** "PMMC is Precise, Meticulous, Mainly Correct – DC only; MI measures More Inclusively (AC+DC)"

Important Constants & Standard Values

Parameter	Typical Value	Machine/Instrument
Transformer Efficiency	96–99%	Power Transformers
Induction Motor Efficiency	85–95%	3-Phase IM
Typical Slip at Full Load	3–8%	Squirrel Cage IM
Supply Frequency (India)	50 Hz	All AC Machines
Winding Factor Kw	0.85–0.96	Alternator
PMMC Accuracy Class	0.1 to 1	PMMC Instruments
MI Accuracy Class	1 to 2.5	MI Instruments
Lamination Thickness	0.3–0.5 mm	Transformer Core
Core Material	CRGO Silicon Steel	Transformers, Motors

Diagram Quick Reference

DC Motor: Armature + Field + Commutator + Brushes + Shaft

DC Generator: Same as Motor + Output Terminals

Transformer: Primary + Core + Secondary + Flux Φ arrows

3-Ph IM: Stator (R,Y,B) + Rotor + RMF arrows

Alternator: Stator + Salient Rotor + Slip Rings + R,Y,B output

PMMC: Magnet (N,S) + Coil + Pointer + Spring

MI: Fixed Coil + Fixed Iron + Moving Iron + Pointer

Wheatstone: Diamond circuit R1,R2,R3,Rx + G + Battery



Best Wishes for Your JNTUK Examination!

Remember: Always draw neat, labeled diagrams. They guarantee marks!

BEEE | JNTUK R23 Regulation | Semester-I | Unit-2: Electrical Machines & Measuring Instruments